

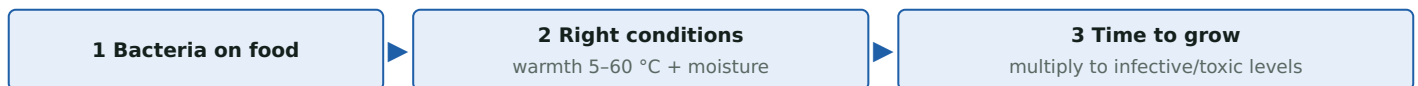
**BEFORE YOU READ — ANSWER FROM MEMORY**

1. What are the 3 links of the "food poisoning chain," and how does breaking one prevent disease?
2. How does an infection differ from an intoxication in timing of symptoms?
3. Which organism's emetic-form toxin is heat-stable and rice-associated, and which produces the most lethal toxin known?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Big picture

- **WHO:** unsafe food causes an estimated **866 million illnesses and 1.5 million deaths** globally each year
- **Definition:** acute gastroenteritis from ingesting food/drink contaminated with pathogenic microorganisms (bacteria, fungi, viruses, protozoa), their **toxins**, or **chemicals**
- **Suspect food poisoning** when acute GI or neurological illness affects **≥2 people who shared a meal in the previous 72 hours**
- Infectious agents: viruses (**Rotavirus, Hepatitis A**), bacteria (**Salmonella, V. cholerae, E. coli, Botulism**), parasites (**Trichinella spiralis**), protozoa (**E. histolytica, Giardia**); also chemicals (pesticides, heavy metals), fungi (mushrooms), and others (**puffer fish / Fugu, Ciguatera**)
- **~68% of foodborne illness and hospitalisation has an unknown agent;** many cases go unreported

2 The food poisoning chain

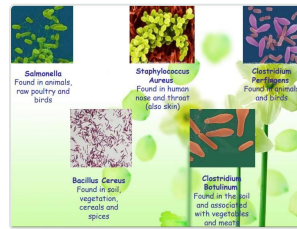
Break any one link and food poisoning is prevented.

Characteristic of food poisoning	
History	Ingestion of a common food ; similar signs/symptoms among those sharing the meal; many people affected at once
At-risk groups	Even the young and healthy are at risk; pregnant women, infants, the elderly, and the immunocompromised are at greater risk
Single cases	Hard to identify unless a distinctive clinical syndrome exists (e.g. botulism)

3 Classification

	Infection	Intoxication
Meaning	Presence of bacteria/microbes that infect the body after consumption	Ingestion of toxins already in the food , including bacterial exotoxins
Onset	Slower — ≥24 hours	Rapid — 2-12 hours

Pathogenesis-based class	Example organisms
1. Preformed toxin ingested	<i>Staphylococcus aureus, Bacillus cereus, Clostridium botulinum, Clostridium perfringens</i>
2. Noninvasive bacteria secreting toxin while adhering to gut wall	Enterotoxigenic <i>E. coli</i> , <i>Vibrio cholerae</i> , <i>Campylobacter jejuni</i>
3. Intracellular invasion of intestinal epithelium	<i>Shigella, Salmonella</i>
4. Bacteria entering the bloodstream via the gut	<i>Salmonella typhi, Listeria monocytogenes</i>



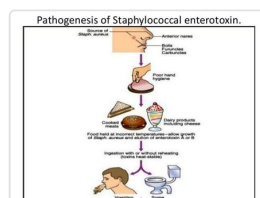
The most common food-poisoning bacteria and their reservoirs.

4 Discriminator grid — the food-poisoning organisms

Organism	Source / reservoir	Onset	Key mechanism
<i>S. aureus</i>	Human nose and throat (also skin); infected food handler	1-6 h	Preformed heat-stable enterotoxin (6 types, A-F) → binds gut receptor → stimulates medullary centre → vomiting
<i>C. perfringens</i>	Animals and birds; soil, dust, sewage; contaminated meat	~ 6 h post-ingestion (toxin-mediated enteritis)	Spores survive cooking → germinate on slow cooling → sporulate in gut → release enterotoxin (type A: alpha + theta toxins) → fluid accumulation
<i>C. botulinum</i>	Soil; associated with vegetables and meats	Variable; fatal course	Preformed neurotoxin , most lethal poison known; blocks acetylcholine release at the neuromuscular junction
<i>B. cereus</i>	Soil, vegetation, cereals, spices (rice)	Emetic 1-6 h / diarrheal 8-16 h	Two enterotoxins: emetic (heat-stable, rice) and diarrheal (heat-labile, meat/milk/vegetables)
<i>Salmonella</i>	Animals, raw poultry, birds, eggs (S. enteritidis in the yolk), rodents	12-72 h	Invades intestinal epithelium → lamina propria → inflammatory mediators, neutrophils, LPS → fever + diarrhea (2-3 days)
<i>Campylobacter jejuni</i>	Poultry, unpasteurised milk	2-11 days (avg 3-5)	Acute inflammation of jejunum/ileum/colon, occasional abscess; fever → foul, watery (sometimes bloody) diarrhea
<i>E. coli</i> O157:H7 (EHEC)	Cattle and other ruminants; uncooked meat, raw milk, contaminated produce	3-8 days	Toxin production once in the gut → bloody diarrhea → HUS ; heat-sensitive, freeze-resistant
<i>Vibrio cholerae</i>	Man is the only natural host; contaminated water, shrimp, vegetables	2 h - 5 days	Multiplies in small intestine → potent enterotoxin → persistent isotonic fluid outpouring

5 Staphylococcus aureus

- S. aureus* is a normal member of the skin / upper respiratory microbiome; **only enterotoxin-producing strains** cause food poisoning
- Food is contaminated by an **infected food handler** (an active lesion can contaminate food); toxin accumulates if food sits at **room temperature**
- 6 heat-stable enterotoxins (A-F)**: bind a bowel receptor, stimulate medullary centres → **vomiting**; toxin/antigenicity **not destroyed by boiling or canning**



Pathogenesis of staphylococcal enterotoxin — nares/skin lesions → poor hygiene → food held at incorrect temperature → vomiting ~6 h later.

6 Clostridium perfringens

- Large Gram-positive rod**; non-motile (usually); non-encapsulated; forms **subterminal oval endospores**; anaerobic; ferments many sugars with gas (**double-zone haemolysis** on blood agar); found in soil, dust, sewage; common in contaminated meat and improperly cooked food
- 4 major toxins**: **alpha** (lecithinase — membrane damage, haemolysis), **theta** (↑ vascular permeability), **kappa** (collagenase — tissue necrosis), **lambda** (protease — tissue destruction)
- Diseases: **food poisoning, gas gangrene (myonecrosis), necrotizing enteritis, cellulitis / wound infection**

- Food poisoning is mainly **type A strains** (alpha + theta toxins) → excessive fluid accumulation in the gut lumen



C. perfringens characteristics and toxins.

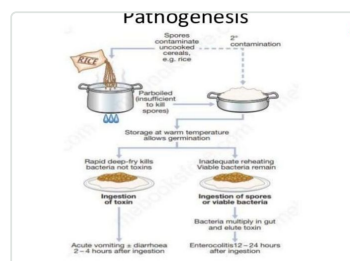
7 Botulism (*Clostridium botulinum*)

WHAT	Neurotoxins , highly toxic; one of the most lethal poisons known — calculated lethal dose for an adult $\approx$ 10 μg ; resists gastric and intestinal juices; not all strains produce toxin
WHY intoxication, not infection	Toxin is preformed in food under anaerobic conditions ; illness follows ingestion of food containing preformed Clostridial toxin. Food heated to 100 °C for a few minutes is rendered safe
MECHANISM	Acts on the parasympathetic nervous system (GI effect is slight); blocks acetylcholine release from the pre-synaptic neuromuscular junction
SIGNS	Dysphagia, diplopia, ptosis, dysarthria , blurred vision, muscle weakness, even quadriplegia ; fever generally retained ; generally fatal ; death in 4-8 days from respiratory or cardiac failure
INFANT BOTULISM	Occurs <6 months of age; infection of the gut by <i>C. botulinum</i> with in vivo toxin production ; toxin absorbed into the bloodstream $\rightarrow$ paralysis

8 *Bacillus cereus*

- **Large motile Gram-positive rod**; ubiquitous in soil and raw, dried, processed foods; **heat-resistant spores** survive cooking, germinate and multiply rapidly at favourable temperature

	Diarrheal form	Emetic form
Toxin	Heat-labile , secreted in intestine	Heat-stable , preformed (like <i>S. aureus</i> toxin)
Incubation	8-16 h	1-6 h
Motility effect	↑ Peristaltic movement	↑ Anti-peristaltic movement
Symptoms	Diarrhea, abdominal pain	Nausea, vomiting, malaise
Duration	12-24 h	6-24 h
Food link	Meat, milk, vegetables, sauces	Rice
Resembles	<i>C. perfringens</i> toxin	<i>S. aureus</i> enterotoxin

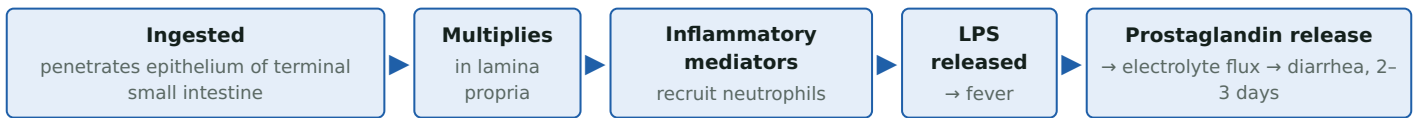


Rice → insufficient parboiling → warm storage allows germination → rapid deep-fry kills bacteria but not toxin (emetic route) vs inadequate reheating leaves viable bacteria (diarrheal route).

9 *Salmonella*

- **>4000 serotypes**; most common: *S. Enteritidis* and *S. Typhi*; source = asymptomatic carriers, contaminated food/water/poultry; route = **faeco-oral**
- Two disease types: **systemic (enteric fever)**: *S. Typhi*, *S. Paratyphi* A-C, invasion continues → generalised infection, little/no diarrhea but pronounced fever; **gastroenteritis (salmonellosis)**: *S. Enteritidis*, infection localises to the ileocaecal region → diarrhea, nausea, vomiting, abdominal pain, temperature may be elevated

- **Killed by pasteurisation**; survive **1 year in moist earth, 16 months in dry earth**; **not destroyed by chilling/freezing** or usual pickling solutions; **S. enteritidis inside the egg yolk**; **convalescent carrier state** can last weeks; **mortality ~1%**; **rats and mice** are a source via urine/faeces



Symptoms usually start 12–72 h after infection and last 4–7 days.

10 *Campylobacter jejuni* and *E. coli* (EHEC / ETEC)

	<i>Campylobacter jejuni</i>	<i>E. coli</i> O157:H7 (EHEC)
Site	Jejunum, ileum, colon — acute inflammation, occasional abscess	Colon; bloody diarrhea
Incubation	2–11 days (avg 3–5)	3–8 days
Clinical course	Preceded by fever → foul-smelling, watery diarrhea 3–4 days, sometimes blood/mucus; abdominal pain with backache; self-limiting, up to 10 days	Stomach cramps, diarrhea (sometimes bloody), fever, vomiting; death rate 3–5%
Complication	—	Haemolytic uraemic syndrome (HUS) — life-threatening
Other <i>E. coli</i> note	—	ETEC = main cause of traveller's diarrhea; mucus + PMNs in stool typical; heat-sensitive, freeze-resistant ; usually self-limiting, 2–3 days

11 Cholera (*Vibrio cholerae*)

- Ingested in food/drink; natural infection dose is **very small**; organism multiplies in the small intestine, producing a **potent enterotoxin** → persistent outpouring of **isotonic fluid**
- **Man is the only natural host**; spread **person-to-person** or via contaminated water/food; **shrimp and vegetables** are frequent carriers; outbreaks occur as **protracted epidemic waves** in endemic areas
- Clinical: frequent vomiting → **rapid dehydration, hypovolaemic shock**; up to **20–30 stools/day**; death in **12–24 h**; extreme dehydration signs — suppressed urine, wrinkled skin, sunken eyeballs, weak husky voice; falling BP, barely audible heart sounds, rapid weak pulse before death

12 Fungal food poisoning

	Detail
Aflatoxins	Produced by <i>Aspergillus flavus</i> and <i>A. parasiticus</i> ; found in rice, potato, peas, maize etc.; 2003 Kenya outbreak — 120 deaths from high-aflatoxin maize; 4 types: B1, B2, G1, G2 ; animals fed B1-contaminated feed secrete aflatoxin M1/M2 in milk ; high-level exposure → acute hepatic necrosis → cirrhosis ; acute hepatic failure: haemorrhage, oedema, altered digestion/absorption/metabolism, mental changes / coma
Mycotoxins (general)	Fungal metabolites; some highly toxic and carcinogenic ; highly stable — not destroyed by normal cooking ; poor dry storage of grains → mould growth → mycotoxin production
Mushroom poisoning (mycetism)	Ranges from mild GI upset to death; usually from misidentifying a toxic wild mushroom as edible (similar colour/morphology); <i>Amanita phalloides</i> is one of the most poisonous toadstools and responsible for most mushroom-poisoning deaths

13 Treatment and prevention

	Detail
Treatment	Rehydration (IV and oral) is the main treatment; avoid solid food while nauseous/vomiting but keep drinking fluids; anti-emetic and anti-diarrheal medication
Prevention	Ultimately depends on educating food handlers about proper cooking, storage and personal hygiene
7 tips	Check best-before dates; throw out doubtful food; cook to the right temperature; refrigerate promptly; clean utensils/surfaces; separate raw from cooked; stay informed

P Must-remember pearls

- Chain = **bacteria + warmth (5-60 °C) + moisture + time**; break any link to prevent disease.
- **Infection** = slow onset (≥ 24 h); **intoxication** = fast onset (2-12 h), preformed toxin.
- ***S. aureus*** (nose/skin, heat-stable toxin) and ***B. cereus* emetic** (rice, heat-stable) mimic each other; ***B. cereus* diarrheal** and ***C. perfringens*** (heat-labile, meat) mimic each other.
- **Botulism** = preformed neurotoxin, blocks ACh release, most lethal poison known (LD ≈ 10 μ g); ***Salmonella*** invades epithelium $\rightarrow$ fever + diarrhea; **cholera** = massive isotonic fluid secretion, death by dehydration in 12-24 h.
- ***E. coli* O157:H7** $\rightarrow$ bloody diarrhea $\rightarrow$ possible **HUS**; **aflatoxins** (*A. flavus*) $\rightarrow$ acute hepatic necrosis $\rightarrow$ cirrhosis.

Q Test yourself

1. A family eats reheated fried rice from the previous night; two hours later they have nausea and vomiting but little diarrhea. Which mechanism is most likely?

- | | |
|---|--|
| A. <i>Bacillus cereus</i> emetic toxin, heat-stable and preformed | B. <i>Salmonella</i> invasion of the ileum |
| C. <i>Vibrio cholerae</i> enterotoxin | D. <i>Campylobacter jejuni</i> invasion of the colon |

2. A food handler with an infected skin lesion prepares a cream cake left at room temperature; guests vomit within 3 hours. Which toxin explains the rapid onset?

- | | |
|--|---|
| A. <i>Salmonella</i> LPS | B. Preformed heat-stable staphylococcal enterotoxin |
| C. <i>Clostridium botulinum</i> neurotoxin | D. <i>E. coli</i> O157:H7 Shiga toxin |

3. Home-canned vegetables are eaten without reheating; the patient develops diplopia, dysphagia and progressive muscle weakness. Diagnosis?

- | | |
|-----------------------|------------------|
| A. Botulism | B. Cholera |
| C. Campylobacteriosis | D. Aflatoxicosis |

4. A child develops bloody diarrhea after eating an undercooked burger, then develops acute kidney injury. Which organism and complication?

- | | |
|--|---|
| A. <i>E. coli</i> O157:H7; haemolytic uraemic syndrome | B. <i>Campylobacter jejuni</i> ; Guillain-Barré |
| C. <i>Vibrio cholerae</i> ; hypovolaemic shock | D. <i>Salmonella</i> ; enteric fever |

5. Which pair correctly matches an organism to its usual food source?

- | | |
|--|---|
| A. <i>Clostridium botulinum</i> — associated with vegetables and meats in soil | B. <i>Staphylococcus aureus</i> — found only in soil |
| C. <i>Campylobacter jejuni</i> — found only in rice | D. <i>Bacillus cereus</i> — found only in raw poultry |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — A** — Rice, short incubation and vomiting-predominant illness point to the heat-stable emetic *B. cereus* toxin.
- 2 — B** — Staphylococcal enterotoxins are preformed and heat-stable, causing rapid vomiting via medullary stimulation.
- 3 — A** — Descending paralysis (diplopia, dysphagia, dysarthria) from a preformed neurotoxin is classic botulism.
- 4 — A** — EHEC O157:H7 causes bloody diarrhea and can progress to HUS.
- 5 — A** — *C. botulinum* is found in soil and associated with vegetables and meats, per the lecture's reservoir figure.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam